

Review of "Linear probing enables near-perfect identification of bird species in acoustic recordings"

Summary

The manuscript presents a two-stage (hybrid) deep learning approach for detecting Eurasian Woodcock (*Scolopax rusticola*) calls in acoustic recordings. The author uses BirdNET embeddings as feature vectors and a secondary classifier is trained, specifically Support Vector Machine (SVM), Random Forest, and XGBoost. The study evaluates the approach on two datasets and reports near-perfect classification performance, with overall average precision scores reaching the values of 99.8%–100% for both cases.

Strengths

1. Clear Motivation and Contribution

The paper addresses a well-defined problem. The performance of the developed model is important for monitoring purposes and biodiversity assessment. The rationale for improving species-specific detection is well-established, given that species-specific classification accuracy can be influenced by environmental noise, overlapping bird calls, and intra-species vocal variations.

2. Well-Constructed Dataset

The author contributes to a valuable annotated dataset. The annotations were created in 3-second intervals, ensuring the coverage of both high-frequency and low-frequency calls of the Eurasian Woodcock (grunts and squeaks). This segmentation strategy guarantees that no part of a call lies outside the annotated intervals, enabling the distinction between positive and negative audio parts. Hence, a strongly labeled dataset is created, in contrast to the weakly labeled training data from Xeno-canto. The dataset has been made publicly available on Zenodo, and as the original audio files are already hosted on Xeno-canto, users can retrieve the source audio data using the provided file numbers, promoting reproducibility.

3. Robust Evaluation Methodology

The cross-validation strategy is thoughtful. From the 2505 negative segments, 5 splits were created as under-sampled subsets to establish a balanced dataset between the positive (501) and negative (501) examples in each fold. Each resulting fold of 1,002 samples (501 positive and 501 negative) is then further divided into 5 train-test partitions, allowing for a cross-validation strategy. This procedure helps mitigate any potential overfitting and provides a reliable assessment of each classifier's generalization capability in distinguishing Woodcock calls from non-target sounds in a variety of acoustic contexts.

4. Comparison with External Studies

The author validates BirdNET baseline performance against literature. Holderried et al. (2025) have also

reported similar values for the performance of BirdNET, i.e. 89.7% true positive rate and 91.9% accuracy for the occurrence of the species, and 81.7% accuracy for the detection of roding events. This indicates that the negative samples in the current study are similar to theirs and that these datasets are both representative for environmental monitoring.

Weaknesses

1. Suspiciously High Performance

The reported results — a mean accuracy value of 0.997 and average precision value of 1.0 for SVM — warrant deeper scrutiny. Near-perfect performance can indicate:

- Data leakage between train and test splits (e.g., segments from the same recording appearing in both).
- Insufficient diversity in negative samples.

The manuscript should describe more clearly whether splits ensure recording-level independence. This is critical, since acoustic segments from the same recording share environmental noise characteristics that could inflate performance.

2. Limited Diversity of Negative Samples

A total of 501 individual Eurasian Woodcock (*Scolopax rusticola*) calls were extracted, along with negative segments from the same recordings where no Woodcock calls were present. Drawing negatives only from recordings that contain the target species may bias the negatives toward similar acoustic environments and could inflate classifier performance. True generalization would require negatives from independent soundscapes.

3. Real-World Generalizability Not Tested

Engler et al. (2025) reports a precision value of 22.0% for BirdNET's performance, i.e. 1475 true positives out of 6803 hits, where their analysis particularly focused on soundscapes near wind farm sites at dusk and dawn times. This dramatic drop in precision in realistic monitoring conditions raises concerns that the 99–100% results reported here may not translate to field deployment. An external validation on a truly independent dataset (different geography, recording equipment, background acoustics) is essential.

4. Figure 2 Provenance

Figure 2 for the linear probing schematic was produced by an AI tool (ChatGPT). While transparency is appreciated, the journal's policy on AI-generated figures should be checked to make sure it is allowed.

5. Minor Issues

- "guaranties" should be "guarantees" in Section 2.1.1.
 - In the legend of Fig. 1: "An example spectrogram of an Eurasian ..."
 - Section 2.1.2: "we concatenated ..."
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Specific Comments

Methods

- **Section 2.3 (Linear probing):** This section is very brief. A more formal description of the mathematical setup would strengthen the paper.

Results

- Confusion matrices would help understand what errors remain in the ~0.2% misclassifications.
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Recommendation

Minor Revision

The paper addresses an important problem and offers a valuable annotated dataset. However, the very high reported performance requires more rigorous validation to rule out data leakage or overfitting to specific recording conditions. Specifically, I recommend the author:

1. **Verify recording-level split independence** and explicitly report the splitting strategy.
2. **Test on an independent held-out dataset** (e.g., recordings from a different year, geography, or equipment).
3. **Report confusion matrices.**
4. **Include hyperparameter tuning details** and ablation on the choice of kernel/classifier settings.
5. **Improve English editing** throughout.

With these revisions, the paper would make a meaningful contribution to automated bioacoustic monitoring.